Scientific Laboratory Report Format

Experiment Title: [Title of Lab Experiment] | Course: Physics / Chemistry / Electrical Lab | Date Performed: [Date]

1. OBJECTIVE & HYPOTHESIS

State the precise experimental purpose. Formulate a testable quantitative hypothesis specifying dependent and independent variables.

2. THEORETICAL BACKGROUND & PRINCIPLES

Explain physical laws, chemical equations, or mathematical theorems governing the experiment (e.g., Ohm's Law, Snell's Law, Beer-Lambert Law).

3. EQUIPMENT & APPARATUS LIST

Catalog of instruments with model numbers, calibration status, and measurement precisions (e.g., Digital Multimeter +/- 0.01V, Spectrophotometer, Vernier Caliper).

4. EXPERIMENTAL PROCEDURE

Step-by-step documentation of experimental setup, parameter changes, safety protocols, and observational checkpoints.

5. OBSERVATION TABLES & RAW DATA

Structured tabular record of observations, repetitions (Trial 1, Trial 2, Trial 3), and arithmetic means.

6. CALCULATIONS AND ERROR ANALYSIS

Show all calculation formulas, percentage error calculation: % Error = |(Experimental - Theoretical) / Theoretical| * 100, and standard deviation.

7. DISCUSSION AND CONCLUSION

Analyze anomalies, evaluate sources of random and systematic errors, and formulate scientific conclusions.